

Machine Learning for teens

By Hamid Sadeghi

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Feature engineering

- ▶ Feature engineering
 - ▶ Making your data easier for machine learning models to understand and learn from.
- ▶ Key Feature Engineering Techniques
 - ▶ Handling Missing Values
 - ▶ Fill in missing data
 - ▶ using mean, median, mode, or predictive methods — or remove
 - ▶ Encoding Categorical Variables
 - ▶ Convert categories into numbers using
 - ▶ Scaling and Normalization
 - ▶ StandardScaler
 - ▶ MinMaxScaler

Feature engineering

- ▶ Key Feature Engineering Techniques
 - ▶ Feature Creation
 - ▶ making new columns (features) from the ones you already have to help your model learn better.
 - ▶ Feature Selection
 - ▶ Choose the most relevant features using
 - ▶ Dimensionality Reduction
 - ▶ Reduce the number of features while keeping important information
 - ▶ Text Feature Extraction
 - ▶ Word2Vec
 - ▶ Bag of Words
 - ▶ TF-IDF
 - ▶ BERT
 - ▶ Outlier Detection
 - ▶ Percentile
 - ▶ Z-score and STD
 - ▶ IQR

Outlier detection and removal using percentile

- ▶ Outliers are data points that are significantly different from others
- ▶ **Percentile**
 - ▶ tells you how a number compares to others.

Standard deviation

- ▶ Standard deviation tells you how spread out the numbers are from the mean.

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

x_i : each value in the dataset

μ : the mean (average) of the dataset

N : total number of values

σ : standard deviation

- ▶ Std
 - ▶ High
 - ▶ the values are spread out far from the average
 - ▶ Low
 - ▶ most values are very close to the average

Z-score

- ▶ Z-score tells you how far a number is from the average, using standard deviation as the unit.

$$Z = \frac{X - \mu}{\sigma}$$

X : the value you're checking

μ : the mean of the dataset

σ : the standard deviation

Z : the Z-score

- ▶ Low Z-score → value is much smaller than average
- ▶ High Z-score → value is much larger than average
- ▶ Z-score near 0 → value is close to average

Assignment

- ▶ I want to use `melb_data.csv`
- ▶ Calculate outlier detection using percentiles, standard deviation, and z-score. Set appropriate upper and lower bounds.
- ▶ Remove the outliers.